

Young's Modulus

How stiff is the *stuff*, as opposed to how stiff is the *thing*

Hang a mass from a steel cable and the cable stretches — a little. Let go and it springs back. So a cable is a spring, and it must have a spring constant k . But k depends on how long the cable is and how thick it is, which are facts about *that particular cable*, not about steel. Young's modulus is the number that describes the steel itself. Everything in this handout follows from separating those two ideas. There is a total of 44 points; problems marked [A] are a little harder.

1 Stress, Strain, and the Definition

Idea 1: Two ratios, and the number connecting them

Take a rod or wire of length ℓ and cross-sectional area A , and pull on it with force F along its length. Define

$$\text{stress } \sigma = \frac{F}{A}, \quad \text{strain } \varepsilon = \frac{\Delta\ell}{\ell}.$$

Stress is “force spread over the area that carries it.” Strain is “fractional stretch” — how much longer, as a fraction of what you started with. For small stretches the two are proportional, and the constant of proportionality is **Young's modulus** Y :

$$\boxed{\sigma = Y\varepsilon} \quad \text{that is,} \quad \boxed{\frac{F}{A} = Y \frac{\Delta\ell}{\ell}}$$

This is Hooke's law, rewritten so that only the *material* appears. Y is a property of steel, or copper, or bone — not of the particular piece you happen to be holding.

Remark

Why divide by A ? Because a rope twice as thick shares the load between twice as much material, so each part of the material is pulled half as hard. Why divide by ℓ ? Because a rope twice as long has twice as much material to stretch, so the same *fractional* stretch produces twice the absolute stretch. Stress and strain are built precisely to cancel out the size of the object.

[3] **Problem 1.** A wire of cross-sectional area $A = 2.0 \text{ mm}^2$ carries a tension of $F = 400 \text{ N}$.

- What is the stress, in pascals?
- If the wire is 1.5 m long and stretches by 0.60 mm, what is the strain?

2 Units, and Why They Are the Units of Pressure

Idea 2: Y is measured in pascals

Strain is a length divided by a length, so it is **dimensionless**. That means $Y = \sigma/\varepsilon$ has exactly the same units as stress:

$$[Y] = \frac{[F]}{[A]} = \frac{\text{N}}{\text{m}^2} = \text{Pa}, \quad \text{and in base units} \quad [Y] = \frac{\text{MLT}^{-2}}{\text{L}^2} = \text{ML}^{-1}\text{T}^{-2}.$$

Those are the dimensions of pressure, and of energy per unit volume — all three are the same thing dimensionally. Real values are large, so Y is usually quoted in gigapascals (1 GPa = 10^9 Pa):

rubber	0.01–0.1 GPa	aluminum	70 GPa
nylon	3 GPa	steel	200 GPa
wood (along the grain)	10 GPa	tungsten	400 GPa
bone	15 GPa	diamond	1000 GPa

- [2] **Problem 2.** Show, using base dimensions M, L, T, that Young's modulus has the same dimensions as energy per unit volume. (This is not a coincidence: stretching a wire stores elastic energy, and Y sets how much.)

Idea 3: An intuition for why the number is so big

Put $\varepsilon = 1$ into $\sigma = Y\varepsilon$. You get $\sigma = Y$. So *Young's modulus is the stress you would need to double the length of the object* — if the material kept behaving linearly all the way there, which of course it never does. Steel snaps at a stress of roughly 0.4 GPa, some 500 times smaller than its $Y = 200$ GPa. Turned around, that says steel breaks at a strain of about $1/500 = 0.2\%$. **Real strains in stiff materials are tiny**, which is why the modulus is huge.

- [3] **Problem 3.** Nylon has $Y \approx 3$ GPa and breaks at a stress of about 75 MPa. At what strain does it break, and how does that compare with steel's 0.2%? Does the more easily broken material stretch more or less before failing?

3 From Young's Modulus to a Spring Constant

Idea 4: $k = YA/\ell$ — the most useful line in this handout

Rearrange $F/A = Y\Delta\ell/\ell$ to put it in the form $F = k\Delta\ell$:

$$F = \frac{YA}{\ell} \Delta\ell \quad \Rightarrow \quad \boxed{k = \frac{YA}{\ell}}$$

Read this carefully, because it is the whole point:

- Y belongs to the **material**. Cut a steel cable in half and both halves still have $Y = 200$ GPa.

- k belongs to the **object**. Cut the cable in half and each half has *twice* the spring constant of the original.
- **Longer** $\Rightarrow$ **floppier**. $k \propto 1/\ell$.
- **Thicker** $\Rightarrow$ **stiffer**. $k \propto A$, so $k \propto r^2$ for a circular wire of radius r .

Remark

You can get both dependences without any algebra, just from how springs combine. A wire of length 2ℓ is two wires of length ℓ joined end to end — springs in **series**, and $1/k_{\text{tot}} = 1/k + 1/k$ gives $k_{\text{tot}} = k/2$. A wire of area $2A$ is two wires of area A lying side by side sharing the load — springs in **parallel**, and $k_{\text{tot}} = k + k = 2k$. Both match $k = YA/\ell$ exactly.

- [4] **Problem 4.** A steel wire ($Y = 200$ GPa) has length $\ell = 2.0$ m and cross-sectional area $A = 1.0$ mm². A 20 kg mass hangs from it. Take $g = 10$ m/s².
- (a) Find the spring constant k of the wire.
- (b) Find how far it stretches.
- [3] **Problem 5.** Two wires are made of the same metal. Wire 2 is three times as long as wire 1 and has twice the radius. By what factor is wire 2's spring constant larger or smaller than wire 1's? By what factor does their Young's modulus differ?
- [3] **Problem 6.** A rubber band and a steel wire of identical dimensions are each stretched by the same *force*. Which stretches further, and by roughly what factor? (Use $Y_{\text{rubber}} \approx 0.05$ GPa.)

4 Oscillations on a Wire

Idea 5: A hanging mass on a wire is just a mass on a spring

Once you know $k = YA/\ell$, vertical oscillations need nothing new:

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{YA}{\ell m}}.$$

Gravity does not appear. It shifts the equilibrium point down by mg/k but does not change the stiffness, exactly as for an ordinary vertical spring.

Example 1: The question you saw on the practice exam

Steel has $Y = 200$ GPa. A mass $m = 300$ kg hangs from a steel wire of circular cross-section with radius $r = 0.15$ cm and length $\ell = 2$ m. Find the frequency of vertical oscillations.

First the area: $r = 0.15$ cm $= 1.5 \times 10^{-3}$ m, so

$$A = \pi r^2 = \pi (1.5 \times 10^{-3})^2 = 7.07 \times 10^{-6} \text{ m}^2.$$

Then the spring constant:

$$k = \frac{YA}{\ell} = \frac{(200 \times 10^9)(7.07 \times 10^{-6})}{2} = 7.07 \times 10^5 \text{ N/m.}$$

Then the frequency:

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{7.07 \times 10^5}{300}} = \frac{1}{2\pi} \sqrt{2360} = \frac{48.6}{2\pi} \approx 7.7 \text{ Hz.}$$

The only step that is specific to this topic is $k = YA/\ell$. Everything after it is ordinary SHM. Watch the centimetres: writing $r = 0.15$ instead of 0.0015 costs a factor of 100 in r , hence 10^4 in A and 100 in f .

- [3] **Problem 7.** In the example above, how far below the wire's unstretched end does the mass hang at equilibrium? Compare that sag with the amplitude of a small oscillation, and comment on whether the wire ever goes slack.

5 Dimensional Analysis with Y

Idea 6: $\sqrt{Y/\rho}$ is a speed

Y has dimensions $\text{ML}^{-1}\text{T}^{-2}$ and density ρ has dimensions ML^{-3} . Divide:

$$\left[\frac{Y}{\rho} \right] = \frac{\text{ML}^{-1}\text{T}^{-2}}{\text{ML}^{-3}} = \text{L}^2\text{T}^{-2},$$

which is a velocity squared. So $\sqrt{Y/\rho}$ is the only speed you can build from a material's stiffness and its density — and it is, up to a factor of order 1, the **speed of sound in that material**. Stiffer material, faster sound; heavier material, slower sound.

- [3] **Problem 8.** Estimate the speed of sound in steel, using $Y = 200 \text{ GPa}$ and $\rho = 8000 \text{ kg/m}^3$. Compare with the speed of sound in air (340 m/s).
- [4] **Problem 9.** [A] A wire of length ℓ , area A , density ρ and Young's modulus Y hangs with a mass m on the end. Without solving anything, use dimensional analysis to argue that the oscillation frequency must have the form $f = \frac{1}{2\pi} \sqrt{\frac{YA}{\ell m}} \times g(\text{dimensionless})$, and identify a dimensionless ratio the leftover function could depend on.

6 Common Traps

Remark

- “**Longer means less stiff**” applies to k , never to Y . A longer rope has a smaller spring constant and the identical Young's modulus. If a problem changes the length and you find yourself changing Y , stop.

- **Watch the area units.** $1 \text{ mm}^2 = 10^{-6} \text{ m}^2$ and $1 \text{ cm}^2 = 10^{-4} \text{ m}^2$. Squaring turns a small slip into a factor of 100 or 10^4 .
- **A radius given in centimetres must be converted before squaring.** This is the single most common way to lose the steel-wire problem.
- **Strain is a fraction, not a length.** $\Delta\ell$ is a length; $\Delta\ell/\ell$ is the strain. Only the second one goes into $\sigma = Y\varepsilon$.
- **Stiff is not the same as strong.** Y tells you how much it stretches under load; the breaking stress tells you when it snaps. Materials can be high in one and low in the other.

7 F = ma Style Questions

Four questions in the style of the exam. Answer each in about two minutes.

- [4] **Problem 10.** A steel wire of length ℓ and cross-sectional area A stretches by 1.0 mm when a certain mass is hung from it. A second wire, made of the same steel with the same cross-sectional area but *twice the length*, carries the same mass. Compared with the first wire, the second wire has
- the same Young's modulus, half the spring constant, and stretches 2.0 mm.
 - half the Young's modulus, half the spring constant, and stretches 2.0 mm.
 - the same Young's modulus, twice the spring constant, and stretches 0.50 mm.
 - the same Young's modulus, the same spring constant, and stretches 1.0 mm.
 - twice the Young's modulus, the same spring constant, and stretches 0.50 mm.
- [4] **Problem 11.** A steel wire ($Y = 200 \text{ GPa}$) of length 2.0 m and cross-sectional area 1.0 mm^2 supports a 20 kg mass. Taking $g = 10 \text{ m/s}^2$, the wire stretches by about
- 0.20 mm
 - 0.50 mm
 - 1.0 mm
 - 2.0 mm
 - 4.0 mm
- [4] **Problem 12.** The speed v of a sound pulse traveling along a thin solid rod depends only on the rod's Young's modulus Y and its density ρ . For steel, $Y = 200 \text{ GPa}$ and $\rho = 8000 \text{ kg/m}^3$. The speed is closest to
- 160 m/s
 - 1600 m/s
 - 5000 m/s
 - 25000 m/s
 - $2.5 \times 10^7 \text{ m/s}$
- [4] **Problem 13.** A mass m hangs from a steel wire of length ℓ and cross-sectional area A , and oscillates vertically with frequency f . The wire is replaced by one of the same steel and the same length, but with *four times* the cross-sectional area. The new frequency is

- (A) $f/4$
- (B) $f/2$
- (C) f
- (D) $2f$
- (E) $4f$

End of handout. If you remember only one line, make it $k = YA/\ell$: the material contributes Y , the object contributes A and ℓ , and everything else is the spring physics you already know.